

KHYATI KIYAWAT

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RESEARCH FOCUS

Interests: Computer Architecture, Processing-in/near-Memory (PIM/PNM), KV Cache Management

Ongoing research: My research focuses on designing and analyzing Processing-in-Memory (PIM) and Processing-near-Memory (PNM) architectures to deliver energy-efficient solutions for GenAI inference workloads.

EDUCATION

University of Virginia, VA, USA

Computer Science Ph.D. Student (*Advised by Prof. Kevin Skadron*)

2022 - present

GPA: 3.88/4.0

Indian Institute of Technology Roorkee, India

B.Tech, Department of Electronics and Communication

2016 - 2020

GPA: 9.23/10.0

INDUSTRY EXPERIENCE

Memory Solution Research Intern | Samsung Semiconductor, USA

May 2026 - Aug 2026

- Mentors: Gongjin Sun, Caroline Kahn
- Characterized the performance of Samsung's CXL-based memory devices and evaluated them as a KV cache offload tier for LLM inference using LMCACHE.

Processor Architecture Research Intern | Intel, India

May 2024 - Jan 2025

- Mentors: Shankar Balachandran, Anant V. Nori, Sreenivas Subramoney
- Designed a framework for tensor tracking in GenAI workloads for architectural exploration on Xeon Processors.

Front-end Design Verification Engineer | Texas Instruments, India Summer 2019, July 2020 - July 2022

- Verified SoC integration of registers, communication peripherals, and co-processors for C2000 real-time MCUs.
- Handled critical silicon debugs for functional test patterns on Automatic Test Equipment (ATE).
- Developed a Perl-based automation framework to auto-generate RTL for memory-mapped registers and the verification test cases thereby significantly reducing manual efforts and errors by 80%.

PUBLICATIONS

- **Reimagining LLM Inference Infrastructure with Memory-Centric KV Cache Servers**
Khyati Kiyawat, Kevin Skadron. *HotInfra*, 2026 (co-located with ISCA 2026). [\[link\]](#)
- **HARMONI: Hierarchical ARchitecture MOdeling for LLMs with Near/In Memory Computing**
Khyati Kiyawat, Yasas Seneviratne, Zhenxing Fan, Morteza Baradaran, Kevin Skadron. *ISPASS*, 2026. [\[link\]](#)
- **Sangam: A Chiplet-Based DRAM-PIM Accelerator with CXL Integration for LLM Inferencing**
Khyati Kiyawat, Zhenxing Fan, Yasas Seneviratne, Morteza Baradaran, Akhil Shekar, Zihan Xia, Mingyu Kang, Kevin Skadron. *arXiv*, Nov 2025. [\[link\]](#)
- **TriPIM — Exact Triangle Counting on UPMEM PIM for Graph Analytics.** (*LBR*) *MemSys*, 2025. [\[link\]](#)
Morteza Baradaran, Khyati Kiyawat, Akhil Shekar, Abdullah Mughrabi, Kevin Skadron.
- **Membrane: Accelerating Database Analytics with DRAM-Based PIM Filtering and Schema Denormalization.** *TACO*, 2025. [\[link\]](#)
Akhil Shekar, Kevin Gaffney, Martin Prammer, Khyati Kiyawat, Lingxi Wu, Helena Caminal, Zhenxing Fan, Yimin Gao, Ashish Venkat, José F. Martínez, Jignesh Patel, Kevin Skadron.
- **Architectural Modeling and Benchmarking for Digital DRAM PIM.** *IISWC*, 2024. [\[link\]](#)
F. Siddique, D. Guo, Z. Fan, M. Gholamrezaei, M. Baradaran, A. Ahmed, H. Abbot, K. Durrer, K. Nandagopal, E. Ermovick, K. Kiyawat, B. Gul, A. Mughrabi, A. Venkat, K. Skadron.

- **An Efficient Scaling-Free Folded Hyperbolic CORDIC Design Using a Novel Low-Complexity Power-of-2 Taylor Series Approximation.** *TVLSI, 2023.* [\[link\]](#)
Anu Verma, **Khyati Kiyawat**, Bishnu P. Das, Pramod K. Meher.
- **Real-Time Minimum Energy Point Tracking Using a Predetermined Optimal Voltage Setting Strategy.** **Khyati Kiyawat**, Yutaka Masuda, Jun Shiomi, Tohru Ishihara. *ISVLSI, 2020.* [\[link\]](#)

RESEARCH PROJECTS

- Scale-free Hyperbolic CORDIC Architecture** | IIT Roorkee, India *Aug. 2019 - Nov. 2021*
- *Advisor:* Prof. Bishnu P Das, ECE Department
 - Designed a low-latency CORDIC architecture to compute sinh and cosh functions with desired precision.
- Power optimization of RISC-V PULPino** | Nagoya University, Japan *Dec. 2019 - Feb. 2020*
- *Advisor:* Prof. Tohru Ishihara, Graduate School of Informatics
 - Proposed a real-time Minimum Energy Point tracking method by analyzing power consumption on RISC-V-based PULPino processor with at most 3.1% energy loss.
- Image Processing based fish seed counting** | IIT Bombay, India *May 2018 - July 2018*
- *Advisor:* Prof. Maryam S Baghini, VLSI Design Lab
 - Proposed a novel data-acquisition apparatus to create a training dataset for a crowd-counting-based CNN model.

TEACHING AND MENTORING

- Graduate Teaching Assistant** | University of Virginia, USA
- Computer Systems and Organization I (CS 2130) - *Fall 2023*
 - Computer Systems and Organization II (CS 3130) - *Spring 2024*
 - CPU/GPU Memory Systems (CS 6501 Special Topics) - *Spring 2025, Spring 2026*
- Graduate Mentorship Programs** | University of Virginia, USA *Fall 2024 - Fall 2025*
- Mentored a junior PhD student on CS doctoral program navigation (CS Graduate Student Group).
 - Mentored a CS undergraduate on graduate school applications (Engineering Grad School Mentor Program).
- Rise Together**  | Charlottesville, Virginia *Oct. 2023 - April 2024*
- Assisted high-school students in the US for college applications and transition process.
- Student Mentorship Programme** | IIT Roorkee, India *Aug. 2018 - May 2020*
- Mentored 12 first-year students in their transition and adjustment to the institute's culture.

AWARDS AND ACHIEVEMENTS

- Selected to participate in *Computing Research Association-Widening Participation (CRA-WP) Grad Cohort Workshop for Women*, San Francisco, USA, 2023.
- Awarded with *UVA Engineering Distinguished Fellowship 2022* given to the top Ph.D. applicants.
- Semi-finalist in *Swadeshi Microprocessor Challenge 2020-21* to build a RISC-V based prototype for autonomous emergency distress system in automobiles.
- Recipient of *Honda Y-E-S Award 2018* awarded to students showing research potential in sustainable technology and also selected for Y-E-S Plus Scholarship 2019 to undergo an internship in Japan.
- Quarter-finalist in India Innovation Challenge Design Contest (IICDC) 2018 organized by Texas Instruments for developing a navigation aid system for fishermen near international waters.
- Awarded with *1988 Batch Award 2017* given to one undergraduate student based on the academic merit.
- Won First prize in Spectrum (circuit design) and Third prize in Ideaz (innovative applications of IoT) at Annual Tech Fest, IIT Roorkee, 2017.

SERVICE

Artifact Evaluation Committee Member | ISCA

2024, 2025

Steering Committee Member | Computer Architecture Student Association - CASA

2025

- Leading the Computer Architecture Long-term Mentorship (CALM) program.

Systems Interest Group (SIG) | University of Virginia

Nov. 2022 - May 2024

- Organized cross-departmental systems research forum, facilitating faculty-student discussions.

Unnat Bharat Abhiyan  | IIT Roorkee

Sep. 2016 - May 2018

- Contributed to the Govt. of India's initiative, leveraging the technical expertise of IITs to drive rural development.

SOPAN | IIT Roorkee, India

Aug. 2018 - May 2020

- Conducted weekly educational outreach sessions for primary and middle school students in a nearby rural village.

PRESENTATIONS

- Presented **Open-Source Processing-In-Memory (PIM) Architecture Design through FPGA Emulation: A Case Study Modeling Sieve** in *Open-Source Computer Architecture Research Workshop*. [\[OSCAR @ISCA 2023\]](#)
- Presented PIMperf: Analytical Modeling for SIMD Instructions on PIM at *CAPWIC 2024*.
- Presented Chiplet-based DRAM PIM Architecture for LLM Inferencing at *TECHCON 2024*.